

In-Depth Analysis of a Retail Accessories Store

An End-Term report for the BDM capstone Project

Submitted by

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Executive Summary

'Sudhir Kumar Belts & Goggles,' is a renowned accessory retailer located on GT Road, Azadpur, New Delhi. Competing among other local vendors, the business operates on a B2C model, delivering good-quality belts and goggles to customers with a strong commitment to excellence. Operating timings of the shop is from 9AM to 9PM.

For the project purposes we are focusing on two major business challenges, one being the issue of dead stock at the store, particularly outdated belts and goggles. This excess inventory ties up capital, occupies valuable storage space, and hinders the introduction of new trends. As a result, it impacts profitability and leads to customer dissatisfaction, making it a critical area for improvement.

The second challenge is the shop's stagnant profit generation, with sales and revenue staying consistent every month. This steady performance prevents the business from leveraging peak seasons or adapting to shifts in market demand, restricting opportunities for growth, innovation, and improvements in overall profitability.

This project aims to provide actionable insights to overcome these obstacles by employing data-driven analysis. Visual tools such graphs, charts, and trend lines were applied to present data and solutions for more clarity.

The analysis revealed that the shop could significantly improve profitability by differentiation between bestselling and other products. Additionally, the business can differentiate itself in a competitive market by offering unique offers on some days. Recommendations include some changes in strategies of selling items and other things.

These insights are expected to help the shop to overcome current challenges, increase market reach, and improve profitability in the near future. Through a more strategic approach, the business can ensure sustained growth and better positioning in the competitive market.

Detailed Explanation of Analysis Process

1. Data Collection -

Data was collected from interviews with the owner of “Sudhir Kumar Belts & Goggles” and various written records such as Khata-book and Daily notepad. Key variables included different brands of both goggles and belts available , cost price & selling price of each item , name of different suppliers and list of specific brand items they supply to the shop , daily and weekly data of number of quantities sold per day of each brand , number of quantities bought & number of quantities left after two weeks.

2. Conversations -

Engaging with the business owner provides qualitative insights that are not captured by quantitative data alone. Understanding the owner’s perspective is also very helpful for me. Additionally, I made regular visits to assess the competitive landscape by mapping the number of accessory shops operating in the same area, giving insight into market saturation.

3. Data Digitization -

All paper-based records, including transaction details, expenses and sales all were digitized. This process created a structured and comprehensive dataset for detailed analysis. Data integrity was ensured through validation checks to minimize errors during entry.

4. Data Cleaning -

The initial data was unstructured and handwritten. Issues included missing values, detecting and removing duplicate records , and unclear records on business operations. Using Excel tools and discussions with the owner the data was cleaned and structured.

I tried to organize the sales and inventory data on both a weekly and daily basis, aiming to simplify the information as much as possible.

WEEK 5 (July29 - August 3)							
Days	SPIKE	PARIS	K-45	2053	2080	6084	RAVYN
Monday	2	0	3	1	0	4	1
Tuesday	3	1	4	0	0	0	1
Wednesday	0	1	1	3	0	1	0
Thursday	2	0	0	0	0	1	0
Friday	5	6	4	3	0	7	0
Saturday	1	2	0	0	0	0	0

Fig 1. Sample sales after data cleaning

5. Descriptive Statistics -

Descriptive statistics helped summarize the data with key metrics such as the average quantities purchased, total cost of purchase, most popular item etc. Identification of high and low-performing items were made, which were visually represented through bar and pie charts, offering clear trends on profit margins and sales.

6. Trend Analysis -

Trend analysis for *Sudhir Kumar Belts & Goggles* involves studying sales patterns to identify customer preferences and seasonal demands. By analyzing trends, the business can stock high-demand items like stylish belts and goggles during peak seasons, reducing dead stock. It also helps in aligning inventory with market preferences, ensuring better profitability and staying competitive in the bustling Azadpur marketplace.

7. Visualization Tools -

Google Sheets and Excel are widely used to organise, clean, and analyse the data. We have used them to make different types of graphs and to do basic calculations to find useful information. Bar charts and pie charts were used to represent the best and worst selling items, customer preferences. Line graphs are used to show inventory data and other things too. Stacked bar graphs are used for comparisons, while trend lines and scatter plots helped to visualise income progressions and the correlations.

8. Weekly Revenue and Expense Tracking -

- Objective: To assess financial performance over time and identify cash flow.
- Method: Weekly revenue and expense data were charted, highlighting discrepancies.

9. Profit Analysis -

Profit Margin : Profit is the surplus amount obtained after deducting the total cost price, rent and other expenses, from the revenue generated. This metric provides a clear indication of the business's financial performance and its ability to generate profits amidst operational costs.

10. ABC Analysis -

ABC Analysis was employed to categorize the inventory based on the value and impact of different items on overall profit generation. I began by extracting data from the shop's purchase and sales records. Items were classified into three categories:

Category A - This category includes high-value items that contribute to roughly 70-80% of the total value but represent a small portion (e.g., 10-20%) of the total inventory.

Category B - This category includes moderately priced items. These items account for about 15-20% of the value and 30-40% of the inventory.

Category C - This category consists of items that contribute to the remaining 5-10% of the value but make up a significant portion (e.g., 40-50%) of the inventory.

“By employing these methodical approaches, the business can gain valuable insights into its financial performance, identify areas for improvement, and make informed decisions to enhance profitability and operational efficiency. This comprehensive analysis lays the foundations for implementing strategic recommendations aimed at optimising pricing strategies, refining selling processes on some specific days, and establishing clear relationship between inventory and sales of shop to drive sustainable growth and competitiveness in the market.”

11. Conclusion of Analysis Process -

The analysis revealed that Sudhir kumar Belts & Goggles performs well in specific market segments, but there are untapped opportunities in some areas. Key recommendations include focusing on products which are slowly becoming a part of dead stock, expanding business in some of the most profitable weekdays ,and differentiating the business through unique services such as various offers and other things. These actions are crucial to stand out in a highly competitive Azadpur market and drive further growth.

Results and Findings

Introduction

This section presents the key insights derived from the data analysis conducted on Sudhir Kumar Belts & Goggles. Through the use of various analytical techniques such as bar charts, pie charts, line graphs, comparison graphs etc. the findings provide a detailed overview of best selling days, space left in inventory, and profitability of each day and many other important things. The results highlight critical trends and opportunities for growth, offering actionable insights that can help the owner to refine their business strategies, increase market reach, and improve overall profitability.

Best Selling Items

To provide solutions or suggestions to the problem statements that I have listed earlier in the previous submissions the first and most basic thing we have to find is which items are best selling and which items are not performing well.

GOGGLES -

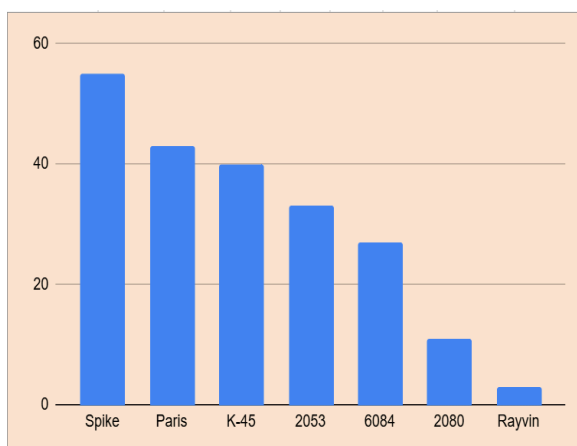


Fig 2. Sales of goggles in the month of July

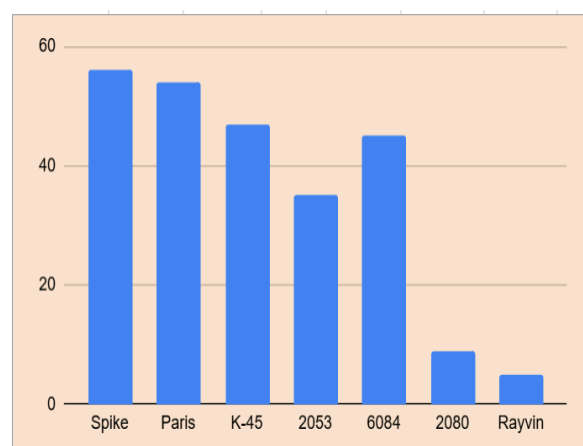


Fig 3. Sales of goggles in the month of August

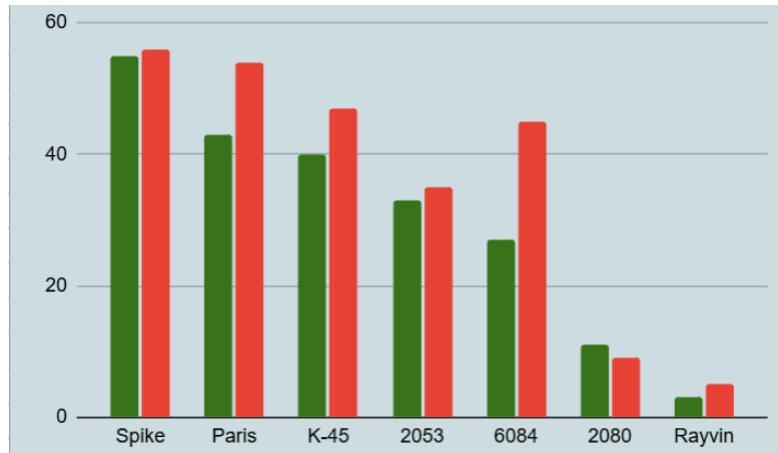


Fig 4. Comparison of sales of goggles in both months

The bar graphs display the sales performance of different goggle varieties for July and August. In both months -

- **Spike** leads with the highest sales, maintaining its dominant position.
- **Paris** follows closely in second place for both months.
- The third top-selling goggle is **K-45**, consistently contributing significantly to total sales.
- **2053** and **6084** show moderate sales, with **6084** slightly outperforming **2053** in August compared to July.
- **2080** and **Rayvin** exhibit the lowest sales figures.

Despite some slight variations in order, the sales hierarchy among the goggle varieties remains generally stable between the two months. The consistency of **Spike, Paris, and K-45** as top performers indicates they are core products driving revenue, while **2080** and **Rayvin** may require promotional efforts or discontinuation strategies to address underperformance. Tracking these patterns can help prioritize inventory management and marketing focus.

The sales data across July and August indicate consistent trends in customer preferences. The similarity in ranking between the two months implies predictable sales patterns, useful for inventory planning and sales forecasting. The significant sales gap between top performers and underperformers suggests the importance of focusing on high-demand products while exploring ways to boost or reevaluate the relevance of slow-moving items.

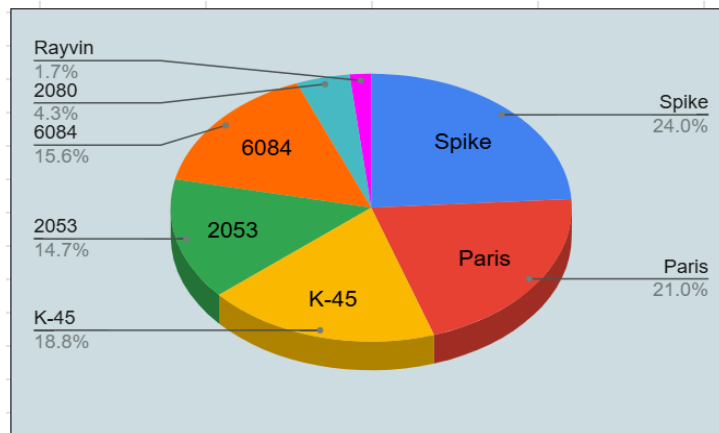


Fig 5. Total share of market sales of different goggles

The pie chart displays the sales distribution of different goggle varieties for July and August. It segments the market share for different goggle types, each marked by a distinct color and labeled with percentages. The variety breakdown is as follows:

- **Spike** : 24.0%
- **Paris** : 21.0%
- **K-45** : 18.8%
- **2053** : 14.7%
- **6084** : 15.6%
- **2080** : 4.3%
- **Rayvin** : 1.7%

Key insights:

- **Top Performer:** The "Spike" goggles variety leads the sales with 24.0%, indicating strong consumer preference.
- **Close Contenders:** "Paris" goggles hold a substantial 21.0% share, closely following the leader.
- **Significant Players:** "K-45" and "6084" varieties also command significant portions of the market, at 18.8% and 15.6% respectively.
- **Moderate Share:** The "2053" goggles hold a notable 14.7% of sales.
- **Minor Share:** "2080" and "Rayvin" varieties have the smallest shares, with 4.3% and 1.7%, respectively.

By pinpointing which varieties are the most and least popular, businesses can adjust their stock and promotional efforts accordingly, ensuring they meet customer demand effectively.

BELTS -

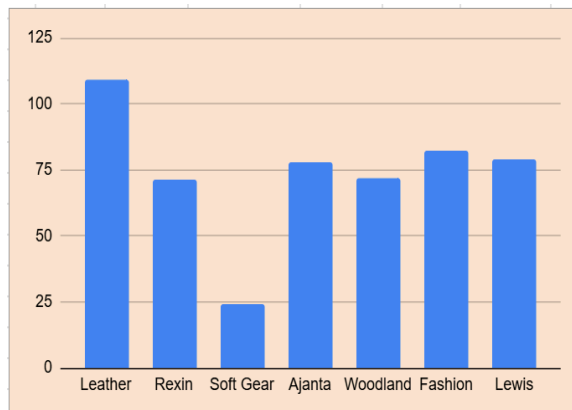


Fig 6. Sales of belts in the month of July

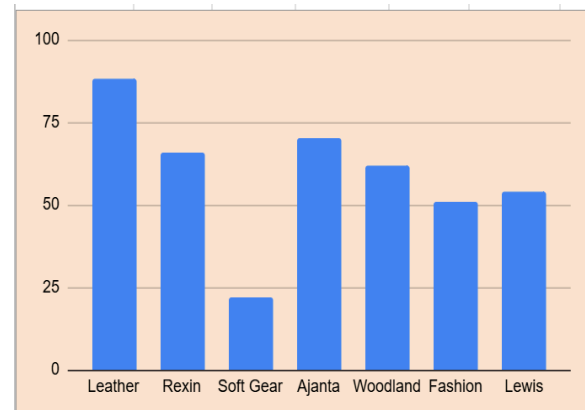


Fig 7. Sales of belts in the month of August

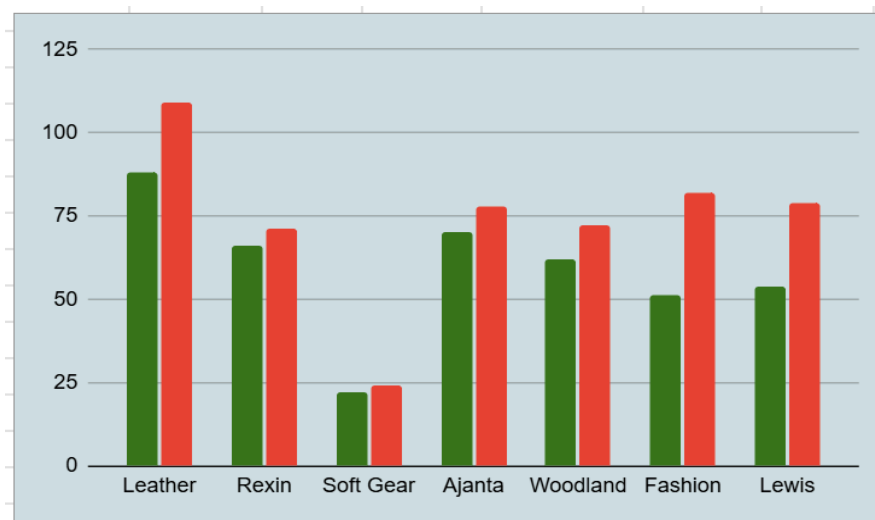


Fig 8. Comparison of sales of goggles in both months

The bar graphs depict the sales of different brands over the months of July (represented by green bars) and August (represented by red bars).

1. **Leather:** There's an increase in sales, rising from approximately 85 units in July to around 115 units in August. This could indicate a growing popularity or a successful

marketing strategy in August.

2. **Rexin:** Sales remained fairly stable with a slight increase from around 55 units in July to about 60 units in August. This stability suggests consistent demand.
3. **Soft Gear:** Sales are the lowest among all brands but showed a minor rise from about 20 units in July to 25 units in August. This suggests a potential but still limited market presence.
4. **Ajanta:** There was a noticeable increase in sales from around 70 units in July to approximately 85 units in August, indicating a positive market response.
5. **Woodland:** Sales increased from around 60 units in July to about 75 units in August. This consistent growth signifies steady demand.
6. **Fashion:** There was a significant boost in sales from around 50 units in July to approximately 80 units in August, suggesting heightened demand or successful promotions.
7. **Lewis:** Sales showed a modest increase from around 55 units in July to about 65 units in August, indicating steady but moderate growth.

In summary, the graphs highlights a general upward trend in belt sales from July to August across most brands, with Leather and Fashion brands experiencing the most significant increases. This overall positive trend suggests an increased demand for belts in August.



Fig 9. Total share of market sales of different belts

Pie chart displays the sales distribution of different belt varieties. It segments the market share for different belt types, each marked by a distinct color and labeled with percentages. The variety breakdown is as follows:

- **Leather** : 21.2%
- **Ajanta** : 15.9%
- **Rexin** : 14.8%
- **Woodland** : 14.4%
- **Lewis** : 14.3%
- **Fashion** : 14.3%
- **Soft Gear** : 5.0%

Key insights:

- **Top Performer:** The "Leather" belts variety leads the sales with 21.2%, indicating strong consumer preference.
- **Close Contenders:** "Ajanta" belts hold a substantial 15.9% share, closely following the leader.
- **Moderate Share:** "Rexin" , "Woodland" , "Lewis" and "Fashion" varieties also command significant portions of the market, at 14.8% , 14.4% , 14.3% and 14.3% respectively.
- **Minor Share:** "Softgear" variety has the smallest share, with 5.0%.

These points highlight consumer preferences and the competitive landscape of the belt market during July and August.

After analyzing the sales trend what I feel is that in belts section most of the belts are performing very good in the market. 6 out of 7 belt varieties are performing above average and only one variety i.e. Soft Gear need some assistance.

Overall, the graph highlights the dominance of Leather belts in the market, while also showing a fairly even distribution of sales among the other varieties, except for Soft Gear. After talking with the owner and some of the customers I got to know that leather belt is the all time favourite for all of them , although the profit margin is not very good in leather belts but it is still very good item for revenue point of view.

Most Productive Days -

For increasing the sales and profit we need to find the most productive days of the week and for this line chart is one of the best tools to show to show the trends.

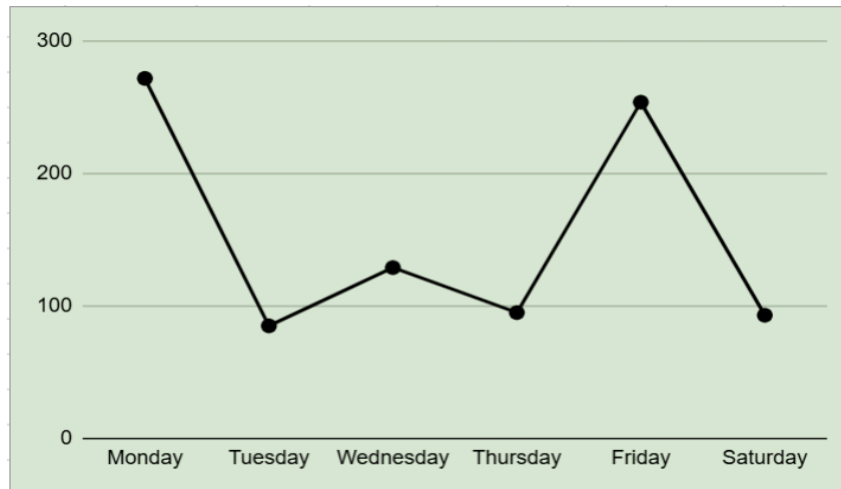


Fig 10. Sales of Belts on each day across two months

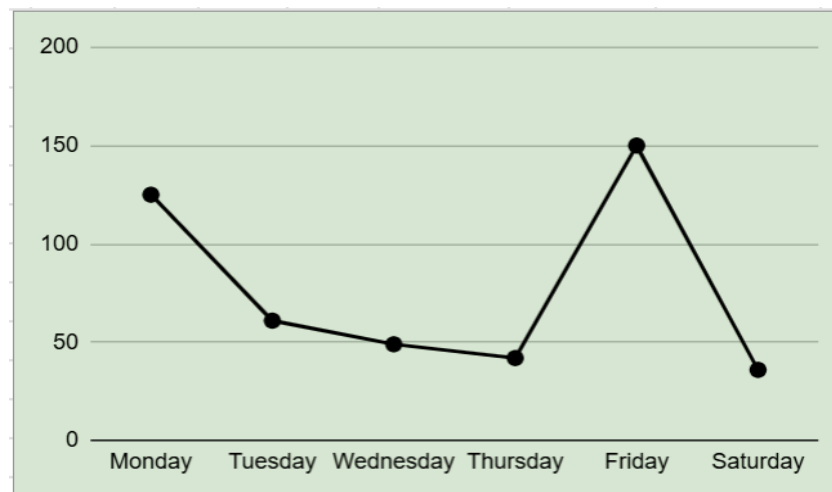


Fig 11. Sales of Goggles on each day across two months

From the above two graphs we can see that in both belts and goggles sales , mondays and fridays are the most productive days with significant differences as compared to other days.

From the graphs representing sales data for belts and goggles across two months, key insights that emerge are -

1. Sales Trends for Belts:

- Monday exhibits the highest belt sales (approximately 300 units), indicating strong customer demand at the start of the week.
- Friday also sees a spike in sales, though slightly lower than Monday.
- Sales decline sharply on Tuesday and remain relatively low until Thursday, reflecting reduced demand mid-week.
- Saturday has the lowest belt sales, suggesting it may be a non-peak day for belt purchases.

2. Sales Trends for Goggles:

- Similar to belts, Friday shows the highest goggle sales, making it a key day for both product categories.
- Sales drop sharply on Tuesday and remain low until Thursday, with Saturday having the least sales.
- The sales pattern for goggles is more consistent but lower in volume compared to belts.

This analysis suggests focusing marketing efforts on peak days (Monday and Friday) and exploring strategies to boost mid-week sales.

After having conversations with owner regarding the rise in sales of products on Mondays and Fridays I got to know that on Mondays, a 'Monday Market' is organised by local market federation on the location where the shop is located, so it benefits the sales.

About Fridays, the owner told that, a 'Friday Market' is organised by same federation about 2 Kms away from the shop, but they go and sell their products in that market and that is the reason for rise in sales on Fridays. This gives a good strategic idea for increasing sales and increasing profits.

Additionally, The significant drop in sales from Tuesday to Thursday suggests a period of reduced customer activity, which presents an opportunity to introduce mid-week promotions or discounts to stimulate demand.

INVENTORY -

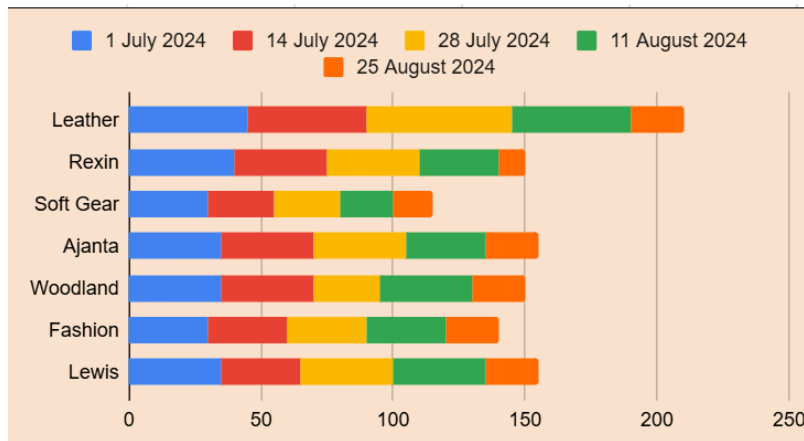


Fig 12. Number of belts added in inventory on particular days

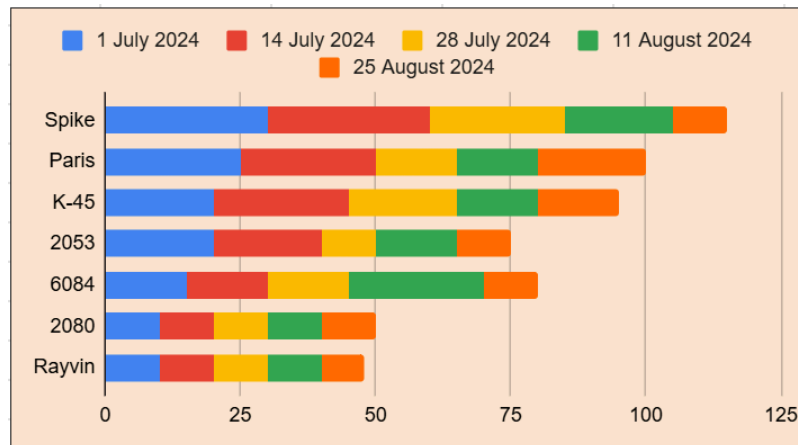


Fig 13. Number of Goggles added in inventory on particular days

During my two-month analysis of the inventory management practices at *Sudhir Kumar Belts & Goggles*, several key insights emerged that highlight both strengths and areas for improvement.

1. Inventory Replenishment Cycle:

The inventory, which includes both belts and goggles, undergoes a complete restock approximately every two weeks. Over the span of my study, this process was observed five times, indicating a consistent and predictable replenishment cycle.

2. Customer Preferences Drive Inventory Choices:

A clear correlation between customer preferences and inventory replenishment was evident. The data shows that popular items such as Leather Belts and Spike & Paris Goggles are stocked in

significantly higher quantities, reflecting their strong market demand. These products have proven to be crowd favorites, consistently contributing to sales and revenue growth.

3. Inefficient Stocking of Low-Performing Items:

Despite the focus on popular products, an inefficiency in inventory management was also identified. Several items with poor market performance are still being purchased in substantial quantities. For example, Soft Gear Belts and 2080 & Rayvin Goggles have demonstrated low customer interest and sluggish sales. Yet, these items continue to be added to the inventory regularly, contributing significantly to the problem of dead stock. This pattern highlights a critical gap in aligning purchasing decisions with actual sales performance, leading to unnecessary capital being tied up in non-moving stock.

Conclusion:

While the business excels at responding to customer demand for high-performing items, an opportunity exists to streamline purchasing decisions for underperforming products.

Implementing a data-driven approach, such as ABC analysis, could help optimize inventory, reduce dead stock, and ultimately enhance profitability. By focusing on the most impactful products and minimizing investment in slow-moving items, *Sudhir Kumar Belts & Goggles* can achieve a more efficient and profitable inventory management system.

ABC Analysis -

ABC analysis categorizes inventory items based on their contribution to overall revenue:

- **Category A:** High-value items with a significant impact on sales.
- **Category B:** Moderate-value items with average sales frequency.
- **Category C:** Low-value items contributing minimally to revenue.

Belts:

- **Category A:** Leather and Rexin belts – These drive the majority of sales and are consistently restocked in large quantities.
- **Category B:** Ajanta and Woodland belts – Moderate sales performance with periodic replenishment.

- **Category C:** Soft Gear belts – Despite low demand, they are frequently restocked, contributing to dead stock.

Goggles:

- **Category A:** Spike and Paris goggles – Highly popular with frequent restocking.
- **Category B:** K-45 and 2053 goggles – Moderate contributors to sales.
- **Category C:** 2080 and Rayvin goggles – Poor performers that add to dead stock.

Optimizing purchase decisions using this classification can minimize dead stock and enhance profitability.

PROFITS -

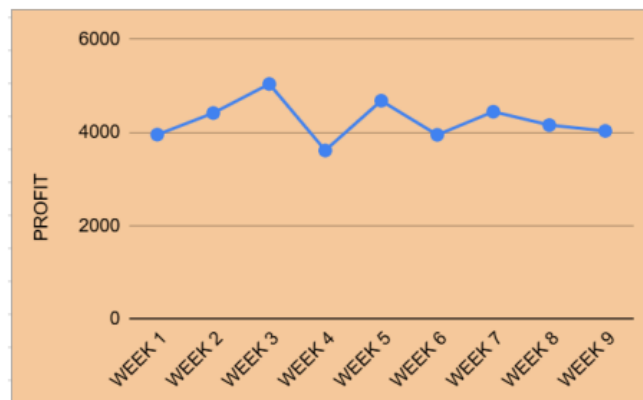


Fig 14. Profit generated per week

This graph depicts the **weekly profit trends** over a span of **nine weeks** for the business *Sudhir Kumar Belts & Goggles*.

Key Observations:

- The profit fluctuates across the weeks but remains generally within the range of **4,000 to 6,000 INR**.
- **Week 3** shows the highest profit, nearing **6,000 INR**, which may correspond to increased sales or reduced operational costs during that period.

- Conversely, **Week 5** exhibits a notable dip, indicating a period of reduced profitability, possibly due to lower sales.
- The profit stabilizes around **4,000 to 5,000 INR** in the remaining weeks, suggesting a relatively steady performance with minor variations.

Conclusion:

As we have already mentioned, The business shows consistent profitability, and the owner also said earlier that the biggest issue that they are facing is stagnant growth.

Some additional Graphs for better understanding -

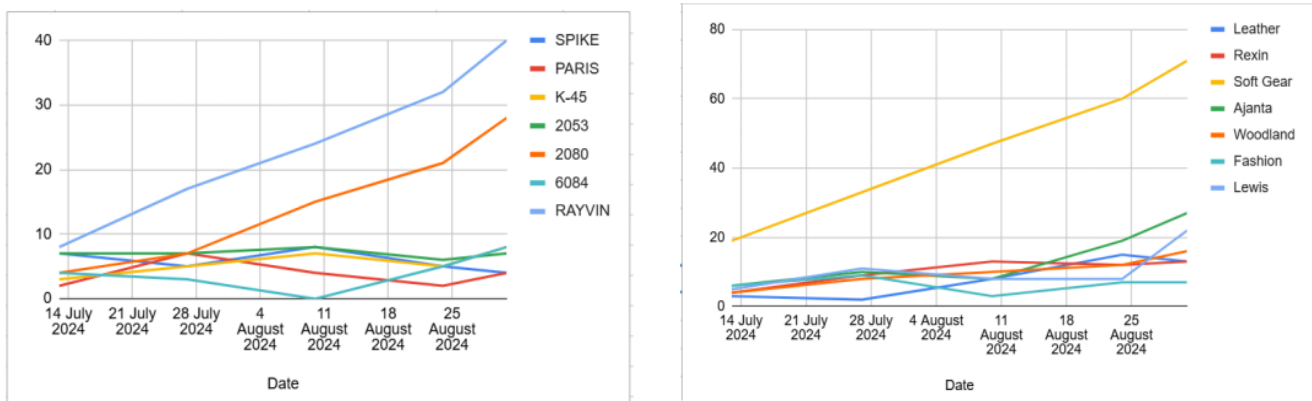


Fig 15. Goggles and Belts left of particular brand in inventory

From the graphs we can see that quantities of Spike and 2080 goggles and Softgear belts are constantly rising in the inventory, which makes them a part of dead stock and reducing the dead stock is very important to increase profit.

Interpretation of Results and Recommendations

1) Recommendations to Increase Profit

1. Focus on High-Margin Products:

Increase marketing and visibility for **Leather belts** and **Spike & Paris goggles**, which are top sellers, to boost their sales further.

2. Bundle Offers and Discounts:

Introduce bundled packages (e.g., a leather belt with trendy goggles) or offer discounts on bulk purchases to drive up average transaction values.

3. Seasonal and Promotional Campaigns:

Utilize festive or seasonal promotions to create urgency and attract more customers.

4. Enhance Store Layout and Display:

Place popular items prominently to maximize customer attention and impulse buying.

5. Leverage Customer Feedback:

Gather and analyze customer preferences to stock new, trendy designs aligned with current market demand.

Recommendations to Reduce Dead Stock

1. Implement ABC Analysis Regularly:

Continuously monitor and categorize items to focus on top-performing products (Category A), control moderate items (Category B), and minimize purchases of low-performing items (Category C).

2. Discount Dead Stock:

Offer deep discounts or clearance sales on **Soft Gear belts** and **2080/Rayvin goggles** to clear out excess inventory.

3. Adopt Data-Driven Purchasing Decisions:

Base inventory restocking on sales performance rather than assumptions to avoid overstocking slow-moving products.

4. **Introduce Limited Stock for Low Demand Items:**

Order low-performing products in smaller quantities to reduce risk and prevent tying up capital.

5. **Return or Exchange with Suppliers:**

Negotiate return or exchange policies with suppliers for unsold or outdated stock to minimize losses.

One of the best solutions is to focus more on MONDAY's & FRIDAY's as those are the market days. On these days giving 2+1 or 3+1 offers would be really great. Offering dead stock items for free with two good performing items would be really helpful to reduce dead stock and increase overall profitability.

ADDITIONAL -

 Collected Dataset